Participate in a Design Thinking for Innovation Workshop, NQF Level 4, Credits 4

PRACTICAL MODULE

Module #	251201-005-00-PM-13:
NQF Level	level 3
Notional hours	40
Credit(s)	Cr 4
Occupational Code	
SAQA QUAL ID	
Qualification Title	

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INTRODUCTION

Congratulations on completing the program. As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility.** Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Acknowledgment of Receipt

I _____ (Learner) acknowledge receipt of my on the job Training Logbook on this the _____ day of _____ 200 ___

The process of on-the-job training has been explained to me.

_____

Signature of Learner

Name of Mentor: _____

_____

Signature of Mentor

Declaration of Authenticity

I (*learner name and surname*) _____, ID Nr _____ hereby declare that the work contained herein was completed by me on my own.

Where assistance or advice was received or where I used resource material form a workbook, policy wording, internet or any other printed sources, this has been acknowledged and referenced. I further declare that I understand that plagiarism is a punishable offence as it constitutes the theft of another's intellectual property rights.

_____ _____

Learner signature Date

Declaration by mentor

I (*mentor name and surname)* _____, ID Nr _____hereby declare that the learner is being mentored by myself and that the functions listed and the working hours is a true reflection of the learner's situation. According to my knowledge I declare that this is his/her own work.

Instructions to the Mentor

Congratulations on your appointment as a mentor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies

- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning
- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the transport operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
- Provide training for mentors and workplace assessors if required and provide the learning material
- Visit employers to monitor progress and provide guidance and feedback.
- Provide an online and telephonic support system to all mentors and learners

Provider Accreditation Requirements for the Module

Provider Accreditation Requirements:

Physical Requirements:

Demonstrate access to:

- Engineering tools
- Work space and bench per learner
- Fabrication material
- Standard operating procedures
- Original Equipment Manufacturer manuals and specifications

Human Resource Requirements:

- Qualified fitter and turner with 3 years of experience in the trade for every 2 learners
- Recognised expertise in mentorship
- Recognised expertise in assessment Legal Requirements:
- Compliance to the relevant occupational health, safety and environmental protection legislation.

- **Additional Assignments to be Assessed Externally**

None specified

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

- 251201005-PM-01, Use Software to Communicate and Visualise Information, NQF Level 4, Credits 3
- 251201005-PM-02, Use and Manage Spreadsheets and Workbooks, NQF Level 4, Credits 3
- 251201005-PM-03, Use Desktop Applications to Analyse, Visualise and report on Data, NQF Level

5, Credits 3

- 251201005-PM-04, Use a Visual Analytics Platform and Visualisation Tools to Analyse, Visualise and report on Data, NQF Level 5, Credits 3
- 251201005-PM-05, Query and Massage Data, NQF Level 5, Credits 3
- 251201005-PM-06, Apply Logical Thinking and Maths, NQF Level 4, Credits 3
- 251201005-PM-07, Apply Code to use a Software Toolkit/Platform in the Field of Study or Employment, NQF Level 5, Credits 3
- 251201005-PM-08, Develop Software using HTML5, Opensource Frameworks and Libraries, NQF Level 5, Credits 16
- 251201005-PM-09, Design and Build Web Applications, Desktop Graphical User Interfaces or Mobile Apps, NQF Level 5, Credits 8
- 251201005-PM-10, Use a Cloud Automation Platform to Create Solutions, NQF Level 4, Credits 8
- 251201005-PM-11, Develop Software using Python, NQF Level 5, Credits 12
- 251201005-PM-12, Apply the Development Cycle when Developing Software, NQF Level 5, Credits 16
- 251201005-PM-13, Participate in a Design Thinking for Innovation Workshop, NQF Level 4, Credits 4
- 251201005-PM-14, Function Ethically and Effectively in a Team, NQF Level 4, Credits 4

Purpose of the Practical Skill Module 10

Purpose of the Practical Skill Modules

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to participate in a design thinking intervention, apply design thinking methodologies and look for opportunities to apply the same methodology in world-of-work and personal life

The learner will be required to:

- PM-13-PS01 : Collaborate with team members to apply innovative and problem-solving strategies
- PM-13-PS02 : Apply design thinking process to solve a problem creatively and innovatively

SECTION A- PM-10-PS01

PM-10-PS01: Collaborate with team members to apply innovative and problem-solving strategies

Applied Knowledge

- AK0101 Design Thinking as a solution-based approach to solving problems

Internal Assessment Criteria

- IAC0101 The ability to collaborate with team members towards a joint goal is demonstrated

INSTRUCTIONS

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

Scope of Practical Skill

Given an engineering training facility fully equipped to deliver practical training on centre lathe, engineering drawings of tapers, the learner must be able to:

- PA0101 Interact and collaborate with team members
- PA0102 Engage in dialogues as required by the process
- PA0103 Participate in the activities as presented by the facilitator
- PA0104 Actively contribute to the discussions and activities

EVALUATION

NAME: …….… **SURNAME:** …….… **COMPANY**…….….. **ID**……...	**EVALUATION CHECKLIST** DATE: TIME:

EVALUATION CRITERIONS	COMPETENT	NOT YET COMPETENT	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA0101 Read and interpret the engineering drawings, and plan and prepare for the turning task			
PA0102 Determine step-by-step operation and the turning sequence of external and internal tapers			
PA0103 Identify and select material, cutting tools and measuring equipment requirements			
PA0104 Calculate and set speeds and feeds for the task			
PA0105 Perform pre-operation inspections on machining equipment and report			
PA0106 Identify potential hazards and risks related to the job and list the			

appropriate responses			
PA0107 Set-up, align and secure the work piece			
PA0108 Set-up cutting tools, coolant and accessories			
PA0109 Sequence the specific turning operation for external and internal tapers			
PA0110 Machine external and internal tapers on a centre lathe			
PA0111 Monitor the lathe while in operation, making adjustments to speeds and feeds where required			
PA0112 Apply quality checks on the machined work piece			
PA0113 Remove, clean and return tools, measuring instruments and accessories			
PA0114 Perform post machining equipment checks and maintenance			

GENERAL COMMENTS OF OBSERVER:		
Date……..	Time started……..	Time completed……
MENTOR/SUPERVISOR NAME ……	**MENTOR/SUPERVISOR SIGNATURE** ……	**LEARNER (SIGNATURE)** ……

SECTION B- PM-10-PS02

PM-10-PS02: Apply design thinking process to solve a problem creatively and innovatively

Applied Knowledge

- AK0201 Involving five phases: Empathize, Define, Ideate, Prototype and Test

Internal Assessment Criteria

- IAC0201 The ability to participate in the design thinking phases is demonstrated
- IAC0202 Skills are used to understand users, challenge assumptions, redefine problems and create

innovative solutions in order to prototype and test an innovative and creative solution

INSTRUCTIONS

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

Scope of Practical Skill

Given an engineering training facility fully equipped to deliver practical training on centre lathe, engineering drawings of external threads, the learner must be able to: •
PA0201 Empathise with users/customers by setting aside own assumptions and formulate real insights about the user

• PA0202 Define the problem statement and create a point-of-view statement, indicating a specific user, a need and an insight

• PA0203 Ideate by challenging assumptions and creating ideas for innovative solutions to help resolve the challenges and needs

• PA0204 Prototype workable ideas

• PA0205 Test the solution applying a reiterative process and using various techniques

Applied Knowledge

• AK0201 Involving five phases: Empathize, Define, Ideate, Prototype and Test

EVALUATION

NAME: ……..… **SURNAME:** ……..… **COMPANY**…….. **ID**……...	**EVALUATION CHECKLIST** DATE: TIME:

EVALUATION CRITERIONS	**COMPETENT**	**NOT YET COMPETENT**	**COMMENTS OR ACTION REQUIRED**
Ability of the learner to:			
PA0201 Read and interpret the engineering drawings, and plan and prepare for the turning task			
PA0202 Determine step-by-step operation and the turning sequence of external threads			
PA0203 Identify and select material, cutting tools and measuring equipment requirements			
PA0204 Calculate and set speeds and feeds for the task			
PA0205 Perform pre-operation inspections on machining equipment and report			
PA0206 Identify potential hazards and risks related to the job and list the			

appropriate responses			
PA0207 Set-up, align and secure the work piece			
PA0208 Set-up cutting tools, coolant and accessories			
PA0209 Sequence the specific turning operation for external threads			
PA0210 Machine external threads on a centre lathe			
PA0211 Monitor the lathe while in operation, making adjustments to speeds and feeds where required			
PA0212 Apply quality checks on the machined work piece			
PA0213 Remove, clean and return tools, measuring instruments and accessories			
PA0214 Perform post machining equipment checks and maintenance			

GENERAL COMMENTS OF OBSERVER:		
Date……..	Time started……..	Time completed……
MENTOR/SUPERVISOR NAME ……	**MENTOR/SUPERVISOR SIGNATURE** ……	**LEARNER (SIGNATURE)** ……

LOGBOOK				
Date	**Assignment task**	**Start**	**Finish**	**Total Hours**
Date	Learner signature	Date	Mentor/supervisor signature	